Shelfie II Project Setup

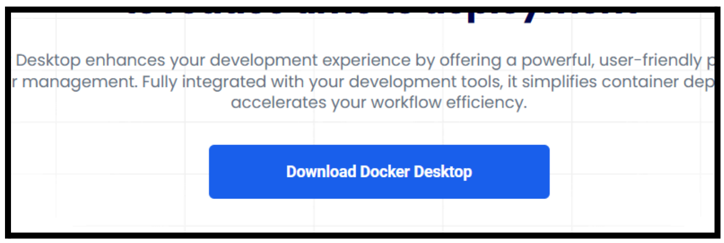
Miami University 2024-2025

With the Shelfie II project there is necessary software needed to run the program. This setup will go through the different software and how to set up the environment that was used for the Shelfie II project. Since the model that we have uses a good deal of Graphics cards resources, the best way to run the code is with some form of Nvidia GPU access. Otherwise it may take a long time to run some of the code or it might not run at all if it is running on your computer's CPU.

Needed software

The needed software for this project is as follows:

1. **Docker**
    a. Docker is a virtual environment software to run a self contained code and make a self contained website on your machine. This software can be downloaded on the Docker website here([Docker]()).



2. **WSL 2 (windows computer only)**
    a. A needed software for Docker to be able to be run on your computer. On Mac or Linux, Docker can run on those systems without this software. For windows the software can be downloaded and installed here([WSL2]()).
3. **VScode or other IDE (optional)**
    a. Is a Code editor that is helpful in changing and seeing the code for the project. It can be downloaded from the link here([vscode]()). This is not required, but may be helpful for people who will want to update or change the code in the future.
4. **SAM**
    a. Is the model used for this project for segmentation of images from META. The model is open source and can be downloaded from the link provided([SAM]()).

Setting up Docker

Once everything is installed from before, based on your system and preference, download the most up to date files from Miami gitlab(link for it I don't know the right link). After that follow these next steps:

1. Get to the file folder in /src/backend/
    a. Should see a compose.yaml file

2. In the folder you will need to add the following
    a. A .env file (will go into more details later)
    b. The SAM model previously downloaded
3. For computers that have Nvidia GPU access follow step 3a otherwise follow step 3b
    a. In the Env file you will want to put (the thing)
    b. For running on a CPU there is a little bit more
        i. The .env file should be left blank
        ii. Then in the compose.yaml file there will be two services uncomment the one for cpu access and comment/remove the one for GPU access
4. Once the compose file and .env file are set up, in terminal in the folder with the compose.yaml file run the command ‘docker compose up --build’
5. After that the container for the docker should be on ‘localhost:8080’ unless changed.

Setup the Frontend

To be added later

Glossary

- GPU (Graphic Processing Unit) -
- CPU (Computer Processing Unit) -
- IDE () -
- SAM (Segment Anything Model) - an open source segmentation model made by META(Example)